
Exploratory Data Analysis (EDA)

Advanced Topics: Methodology, Philosophy, and Research Design

1. What is EDA?

Exploratory Data Analysis (John Tukey 1977): Discovering data before building models.

Philosophy: 'It is important to understand what you CAN DO before you learn what you are SUPPOSED TO DO.' (Tukey).

EDA prioritizes graphical representation over formal tests. Four goals: understand data, find patterns, detect anomalies, check assumptions.

NOT: hypothesis testing. EDA generates hypotheses for independent data testing.

2. Measures of Central Tendency and Dispersion

Mean: arithmetic average. Sensitive to outliers.

Median: 50th percentile. Robust to outliers.

Standard deviation (SD): average deviation from mean. Squared units.

Interquartile range (IQR): $Q3 - Q1$. Robust spread indicator.

MAD: Median Absolute Deviation. More robust than SD and IQR.

3. Graphical Methods

Histogram: Distribution of one variable. Bin width matters: Sturges rule or Freedman-Diaconis.

Boxplot: Median, IQR, $1.5 \times \text{IQR}$ whiskers, outliers.

Violin plot: Boxplot + kernel density estimate. Shows more detail (bimodality).

Scatterplot: Relationship between two metric variables.

Q-Q plot: Data quantiles vs. theoretical distribution quantiles.

4. Outliers and Influential Points

Outlier: Value far from distribution center. Detection: $> 3 \times \text{IQR}$ from median, or $|z| > 3$.

Influential point: Single case that strongly affects the model. Detection: Cook's distance.

Handling: Do not automatically delete! Is it measurement error? Legitimate variation? New phenomenon?